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Sustainable Fuel Management in Mining: Fuel Optimization Strategies in Loading and Haulage

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ABSTRACT

Mining industry is considered as one of the major energy-consuming sectors. Fuel optimization can contribute to sustainable mining through saving non-renewable energy resources as well as mitigation of environmental impacts of greenhouse gas (GHG) emissions. Since, loading and haulage account for a considerable amount of energy consumption in mining activities, much attention has been paid to optimize fuel consumption in these operations. This paper offers a concise exploration of fuel consumption optimization strategies in mining with a specific focus on loading and haulage operation, including challenges and limitations. For this purpose, the study investigates fuel consumption factors in loading and haulage, examines strategies, and discusses future directions while considering inherent challenges. The review underscores the notable emphasis on fuel-saving strategies within the realm of mining short-term planning. In this context, this paper outlines the effects of implementing our monitoring and dispatching system in a large-scale mine. This resulted in a notable 9.7% reduction in fuel consumption, consequently leading to a substantial decrease of approximately 3 million kilograms in CO_2 emissions for the span of a yea. Furthermore, alternative solutions, such as In-Pit Crushing and Conveying, and trolley assist system, as well as the exploration of alternative fuels like natural gas and hydrogen, are advocated for optimizing fuel consumption and diminishing greenhouse gas emissions. Moreover, there is a focus on strategies examining maintenance-related aspects to further enhance fuel optimization.

Keywords: Sustainable mining; fossil fuel; optimization; loading and haulage.

1. Introduction

Nowadays, the mining sector needs to confront diverse challenges, including decreasing ore grades, increasing mining depth (resulting in longer haulage distances and higher transport costs), escalating energy consumption, and growing environmental concerns. To address these issues, the mining industry should integrate the principles of green mining into its operations. The concept of green mining is to improve the mining industry holistically so that it is safe, efficient, and environmentally sustainable (Zhou et al., 2020) and can be achieved by balancing its three components, including economic, social, and environmental aspects (Muduli and Barve, 2011). Over the years, considerable attention has been devoted to the optimization of mining production systems and enhancing energy efficiency in mining as strategic initiatives contributing to sustainable mining practices.

Among mining activities, loading and haulage account for a significant proportion of energy usage. Considering that a major part of this energy is derived from fossil fuels, this paper presents a comprehensive review of strategies to optimize fossil fuel consumption in mining operations. Despite the growing advancement of renewable energy sources, non-renewable fossil fuels such as oil, coal, and natural gas continue to account for over 80% of the world's total energy supply (Holmberg et al., 2017). The total amount of energy used in the mining and minerals industry has been estimated to be 4–7% of the global energy output (Holmberg et al., 2017). Loading and haulage operations stand as significant fuel consumers within the mining operation. Fuel consumption of mining dump trucks accounts for about 30% of total energy use in surface mines (Siami-Irdemoosa and Dindarloo, 2015). As an illustration, fuel usage reports from the Bistun Kavir Yazd intelligent mining system reveal that, over the course of a month, haulage and loading operations in a specific iron ore mine constituted 69% and 21% of the total fuel consumption, respectively, as shown in Fig. 1. It is worth noting that these data pertain to the extraction operations (from drilling to the start of comminution) and, additionally, a portion of drilling and loading operations' energy in this mine is provided through electrical energy.

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Prioritizing energy conservation in loading and haulage operations is crucial as it not only lowers energy consumption in mining but also brings financial benefits and reduces environmental impacts. By incorporating the principles of green mining, wherein optimizing fossil fuel consumption holds paramount importance, the mining industry can make significant strides towards enhancing its sustainability while effectively addressing the current challenges.

Accordingly, this paper focuses on providing a comprehensive review of fuel consumption optimization strategies in loading and haulage operations. Recognizing the paramount importance of optimizing fossil fuel usage as a key aspect of green mining, we examine various methods and approaches to enhance fuel efficiency in these essential processes. Our aim is to offer valuable insights that contribute to sustainable and environmentally responsible practices in the mining industry, with an emphasis on fuel consumption optimization in loading and haulage activities.

With this objective in mind, in the subsequent sections following the literature review, we will analyze the key factors influencing fuel consumption. Subsequently, we will introduce a comprehensive range of strategies and techniques aimed at enhancing fuel efficiency. In this context, we will highlight the practical application of optimization through Monitoring and Dispatching Systems. Furthermore, an examination of the challenges and limitations associated with these approaches will be undertaken, along with a discussion of future directions.

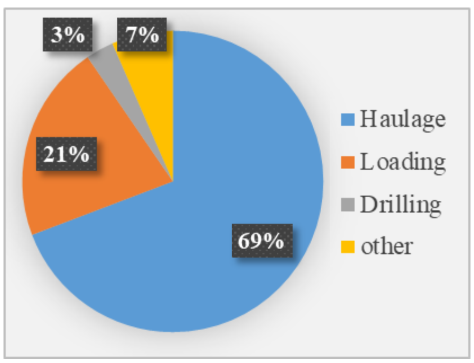

Fig. 1. Monthly fuel breakdown for a specific mine (from drilling to the start of comminution - excluding electrical energy used for a portion of loading and drilling operations for a clear representation of fuel consumption) - Data source: Bistun Kavir Yazd intelligent mining system

2. Literature review

Numerous scholarly papers have extensively investigated the field of fuel energy conservation in mining, due to its compelling financial advantages, such as cost savings and improved profitability. Additionally, it leads to positive environmental outcomes including lower greenhouse gas emissions and fewer environmental impacts. Cooke and Randall (1995) found in their investigation on energy use benchmarks for open-cut coal mines that emphasizing greater electricity utilization across mining operations resulted in reduced overall energy costs, taking advantage of the pricing disparity between electricity and diesel at the time of the study. Kecojevic and Komljenovic (2010) examined the correlation between a truck's fuel consumption, power, and engine load factors, alongside evaluating the quantity of CO_2 emissions and the corresponding costs that might arise due to potential CO_2 legislation. Norgate and Haque (2010) conducted a comprehensive assessment of the lifecycle of mining and mineral processing for iron ore, bauxite, and copper concentrate, focusing on evaluating embodied energy and greenhouse gas emissions. They determined that loading and hauling constitute the most substantial contributions to overall greenhouse gas emissions in the mining and processing of iron ore and bauxite. Additionally, for the production of copper concentrate from copper ore, the crushing and grinding stages accounted for the largest portion of total greenhouse gas emissions. Paraszczak et al. (2014) investigated electric drives, mainly those commercially available, as viable alternatives to diesel power trains. They discussed the advantages of electric drives and explored their potential implementation in the unique conditions of deep metal mines. Baidya et al. (2019) examined the viability of recuperating waste heat from diesel generator exhausts, revealing prospects for energy and economic savings, thus enhancing sustainability and cost-effectiveness in mining operations. Golbasi and Kina (2022) conducted an evaluation of kinematic fuel consumption factors of haul trucks operating simultaneously within a multi-route operation network under stochastic payload and precipitation conditions. Their findings revealed that precipitation conditions could result in a fuel consumption variation of approximately 15–25 percent. Jamaković and Mašić (2022) conducted an extensive analysis of fuel consumption monitoring data collected over a 6-month period to determine the amount of carbon dioxide emitted into the atmosphere by the BelAz dump trucks. Feng et al. (2022) examined heavy-duty mining haul trucks demonstrated that battery electric alternatives offer both cost-effectiveness and environmental benefits, while recognizing the need for ongoing technological advancements to address challenges associated with hydrogen fuel cells. Meneses and Sepúlveda (2023) investigated the effects of temporal deterioration on mining roads in an open pit operation, evaluating productivity and fuel consumption through discrete event simulation. Their findings underscored that overlooking road conditions could lead to overestimating productivity by up to 600 Ton/hr and increasing fuel consumption by 78%.

In summation, the extensive literature underscores the vital importance of fuel energy conservation in mining. Previous studies reveal a range of benefits, from financial advantages like cost savings to positive environmental outcomes such as reduced emissions. These findings pave the way for our in-depth exploration of fuel optimization strategies in loading and haulage operations, contributing to the drive for sustainable mining practices.

3. Influential factors affecting fuel consumption

In the context of a comprehensive review of pertinent literature examining loading and haulage operations, it becomes apparent that the determinants influencing energy efficiency can be systematically categorized into four distinct classes: configuration of equipment, operational circumstances, mine design and plan, and proficiency and experience of human resources. Optimizing these variables contributes to the enhancement of loading and haulage operations, ultimately resulting in improved fuel consumption. A selection of relevant literature in this context is outlined in Table 1, followed by comprehensive elucidations of loading and haulage operations in the ensuing subsections.

Table 1. Literature on Fuel Consumption Optimization in Loading and Haulage Operations

Loading		
Factor(s)	Results	References
Blast design	12-46% better total dig time with P80 fragmentation change from 600 to 200 mm	(Brunton et al., 2003)
Attributes of muckpile	Higher mean particle size and index of uniformity lead to decreased bucket fill factor and production rate	(Singh & Narendrula, 2006)
Operating Condition (material digability)	7–58% dig time enhancement from muck pile and operator	(Soofastaei et al., 2018)
Operator, material digability, bench height	Operator experience contributes significantly	(Bettens et al., 2022)
Haulage		
Factor(s)	Results	References
Payload variance	Savings of 35% in total fuel and GHG gas emissions cost	(Soofastaei et al., 2016a)
Operator proficiency	Operators significantly impact loading & hauling energy efficiency	(Awuah-Offei, 2016)
Truck bunching (due to Payload variability)	Higher resistance and payload variance increase fuel consumption. Increased truck capacity reduces consumption	(Soofastaei et al., 2016b)
Mine design	14% reduction in fuel consumption - 16% increase in productivity	(Rodovalho & de Tomi, 2016)
Gross vehicle weight, truck velocity, total resistance	Max velocity, total resistance, and gross weight influence fuel use: 60, 26, and 14 relative importance scores	(Soofastaei et al., 2016)
Driving style	Aggressive driving consumes 10% more fuel	(Wang et al., 2020)
Equipment, Staff, operating times	Fuel consumption model considering energy factors	(Yousefi Nejad et al., 2021)
Lifting, distance, speed, load, time, age, type	Lifting height, load, and haulage distance greatly impact fuel consumption, while queuing time also plays a role	(Wang et al., 2021)
Payload, traffic, queueing, and maintenance conditions	Lower truck fuel emissions by over 30% without impacting production levels	(Huo et al., 2023)

3.1. Loading operation

An optimized loading process contributes to heightened production efficiency, enhanced energy conservation, and decreased operational costs. Within the scope of loading operations, a significant portion of studies has centered on the impact of the four mentioned parameters (equipment, operating condition, mine design, and operator skill) on shovel productivity.

Various factors affect the productivity of loading machines. The distribution of blast fragmentation sizes and the design of the blasting process directly affect the load and haul cycle by influencing excavator dig time and bucket payload (Brunton et al., 2003). In addition, the efficiency of loading is significantly influenced by the attributes of the material being loaded (Singh & Narendrula, 2006). Looseness in the muck affects both the fill factor and the cycle time of a loader, subsequently affecting equipment productivity. Moisture content alters the material's angle of repose and its stickiness to the loader's bucket. Studies indicated that larger mean particle sizes result in reduced bucket fill factor and production rate. Moreover, the design of the loading machine significantly impacts its productivity. Specifically, in the case of shovels, three factors can enhance cycle time: increased pull torque, higher peak power, and enhanced swing and dipper movement speed (Soofastaei et al., 2018). Furthermore, the skill and experience of the operator significantly influence the productivity of loading machines. Ultimately, a harmonious alignment between bench height and the machine's capabilities contributes to enhanced productivity (Bettens et al., 2022).

3.2. Haulage operation

In the context of haulage operations, there exist parameters that significantly influence efficiency. Understanding these factors holds the potential to enhance fuel consumption optimization. These parameters encompass various factors, including operational conditions (e.g., payload variance, truck bunching, gross vehicle weight, truck velocity, haulage fleet operating time), workforce elements (operator skill, driving style), mine planning and design (equipment count, loading-unloading distance, haulage route grades), and equipment characteristics (type and age).

Information derived from payload management systems of haul trucks across different surface mines highlights the significant payload variability (as one of the elements associated with operational circumstances) that must be taken into account when analyzing mine productivity, diesel energy consumption, greenhouse gas emissions, and associated costs (Soofastaei et al., 2016a). A range of factors, including particle size distribution, swell factors, material density, truck-shovel match, shovel passes, and bucket fill factor, influence payload variance (as cited in (Soofastaei et al., 2016a)). Similarly, among the other factors in this category (operating condition), Gross vehicle weight, truck velocity, and total resistance (the summation of rolling resistance and grade resistance) are acknowledged as the important parameters influencing fuel consumption (Soofastaei et al., 2016). It is worth mentioning that rolling resistance is influenced by tire and road surface properties, with some aspects linked to machine design. Moreover, lifting height, truck speed (load and empty), and operating times (loading, travel, dumping, and idle states) are discussed as significant factors in the literature (Yousefi Nejad et al., 2021; Wang et al., 2021).

The influence of operators is another contributing factor to energy efficiency, and the potential of operator training as a viable strategy for improving energy efficiency should not be overlooked. There is both theoretical (modeling) and empirical evidence indicating that operators can exert a substantial influence on the energy efficiency of loading and hauling operations (Awuah-Offei, 2016).

Another important cluster of factors that significantly influence operational productivity and fuel consumption pertains to the realm of mine design and planning. The integration of efficient mine design can contribute to both increased production and decreased fuel consumption, mitigating the environmental impact of mining activities (Rodovalho and de Tomi, 2016).

Ultimately, it is essential to underscore the significant influence of equipment characteristics on fuel consumption. Factors encompassing equipment type, age, and design intricacies contribute to the efficiency of fuel usage in mining operations. Numerous scholarly works have explored these parameters, alongside those mentioned earlier, as influential factors in fuel consumption, delving into the relationship between equipment attributes and the optimization of fuel usage (Yousefi Nejad et al., 2021; Wang et al., 2021).

4. Optimization strategies for loading and haulage operations

Upon examining the existing literature, it becomes evident that various methods and techniques aimed at optimizing fossil fuel consumption can be categorized under specific strategies. These encompass optimizing short-term planning, integrating fossil fuel alternatives, and enhancing maintenance programs. This section will expound upon each of these strategies in detail.

4.1. Short-term planning

Short-term and operational plans are designed to address operational specific details, ensuring coherence with medium and long-term objectives, while effectively achieving daily, weekly, or monthly operational goals. This approach is crucial for ensuring the sustainability of mining operations (Monika and Paszkowska, 2004) and is considered essential for meeting the operational objectives and strategic goals of the mine (Upadhyay and Askari-Nasab, 2018).

General solutions for achieving sustainable management in mine operations encompass enhancing operational efficiency, embracing state-of-the-art technologies, and implementing digitization. The pursuit of sustainability within short-term planning involves guaranteeing appropriate mine design, optimizing sequencing, and achieving precise time forecasting in mine operational tasks (Rahnema et al., 2023). To address environmental challenges and promote energy efficiency, short-term planning presents diverse possibilities, including optimizing machine idle periods, mitigating traffic congestion, prioritizing excavations, allocating fleets efficiently, minimizing redundant material handling, and relocating machinery. These opportunities have the potential to enhance energy efficiency throughout the equipment's lifespan and lower greenhouse gas emissions.

As a strategy to enhance loading and haulage productivity and consequently improving energy efficiency, significant attention has been directed towards optimizing truck-shovel allocation problems in recent years. A diverse range of techniques, including mathematical modeling (Bajany et al., 2017; Mirzaei Nasirabad et al., 2023), simulations (Bajany et al., 2019; Awuah-Offei et al., 2012; Soofastaei et al., 2015b; Dumakor-Dupey et al., 2017), and artificial intelligence techniques (Soofastaei et al., 2016), are employed to optimize the allocation of trucks to shovels.

The optimization within this scope involves the evaluation of influential factors affecting fuel consumption, including but not limited to operational durations, haulage distances, dump truck speeds, and the quantity of trucks. In addition, numerous endeavors have been undertaken to decrease payload variance through the adoption of cutting-edge technologies. These efforts involve the incorporation of truck onboard payload measurement and the utilization of real-time fleet monitoring systems (Knights and Paton, 2010; Hewavisenthi et al., 2011). Our literature review demonstrates fuel savings ranging from 4 to 17 percent attained through the utilization of these techniques (Bajany et al., 2017), (Sahoo et al., 2014).

4.2. Alternative solutions

The exploration of alternatives to fossil fuels is crucial given environmental concerns, resource limitations, and the necessity to tackle climate change. Furthermore, within the context of the global shift towards decarbonization, it is essential to investigate strategies for reducing carbon emissions in haulage systems (Bao et al., 2023).

In the domain of emerging technological possibilities, electrical alternatives hold the promise of decreasing energy consumption, thereby contributing to mining systems with a lower carbon footprint. In this context, scholarly literature has directed its attention towards systems such as In-Pit Crushing and Conveying (Nehring et al., 2018; Purhamadani et al., 2021), innovative Trolley Assist (TA) (Bao et al., 2023; Cruzat and Valenzuela, 2017), Battery Trolley systems (Bao et al., 2023), and electric load-haul-dump units (Jacobs et al., 2015). In the current global scenario of escalating energy expenses, continuous mining solutions like In-Pit Crushing and Conveying (IPCC) systems have emerged as a genuine substitute for conventional truck-based haulage systems. Trolley Assist solutions have exhibited impressive potential in reducing emissions within their operational contexts. Developments in battery and energy recovery technologies are shaping the evolution of TAs, transitioning them from diesel-electric truck-based setups to fully electrified Battery Trolley systems. The integration of Battery Trolley systems alongside autonomous battery-electric trucks and Energy Recovery Systems (ERSs) represents an innovative approach that holds the potential for substantial emission reductions in surface mining operations. This approach is aligned with safety enhancements, energy conservation, and operational advancements (Bao et al., 2023).

Moreover, in an effort to identify substitutes for conventional fossil fuels, various solutions are exploring alternative options such as Natural Gas (NG) (Feng and Dong, 2019), hydrogen (Figueiredo et al., 2023; Benbellil et al., 2022), and Hydrotreated Vegetable Oil (HVO) (Aatola et al., 2008). Natural gas (NG) emerges as a promising substitute fuel to replace diesel in heavy-duty vehicles, exhibiting significant potential for reducing fuel costs and emissions (Feng and Dong, 2019). Nevertheless, employing hydrogen as a fuel encounters challenges arising from complexities in production, storage, transportation, and supply. However, it can serve as an additive to fossil fuels, rendering it a feasible alternative (Figueiredo et al., 2023). Hydrogen emerges as a prime contender for diesel enhancement due to its renewable nature, alignment with engine specifications, performance augmentation, and significant reduction in carbon emissions (Benbellil et al., 2022).

4.3. Maintenance

An appropriate maintenance plan has potential to reduce energy consumption of the mining fleet (Peralta et al., 2016; Vera-Burau et al., 2023). Through the adoption of a comprehensive maintenance policy, a range of energy-saving advantages can be realized, as supported by the literature. A well-structured maintenance plan can enhance equipment reliability, a critical factor influencing both energy consumption and greenhouse gas emissions (Peralta et al., 2016). Through the reduction of maintenance intervals, the ability to elevate tire pressure emerges, leading to a reduction in rolling resistance and consequent fuel consumption decrease in haul trucks (Soofastaei et al., 2015a). Nevertheless, this approach may lead to increased maintenance costs, and it is essential to establish a maintenance threshold to ensure cost-effectiveness.

4.4. Fuel optimization via monitoring and dispatching systems: practical application

As an exemplification of fuel optimization through the implementation of monitoring and dispatching systems, we present a specific case study in this context. In one large-scale mine, we successfully deployed the state-of-the-art intelligent mining system, BISMART, developed by Bistun Kavir Yazd Company.

BISMART, as an intelligent system, actively engages in optimizing mining operations through three key strategies. Firstly, the system excels in monitoring mining activities, enabling swift responses to production obstacles. Continuous tracking of travel routes ensures that any issues in specific sections are addressed promptly, minimizing downtime and enhancing overall productivity. This approach significantly contributes to reducing operational disruptions and increasing the efficiency of mining processes.

Secondly, it employs advanced strategies for allocation of trucks to loaders, playing a pivotal role in operational optimization. The system utilizes both fixed and flexible allocation approaches, allowing for customization based on the unique conditions of each mining site. Through the implementation of allocation algorithms, the system guarantees the assignment of trucks to loaders in a manner meticulously designed to minimize operational durations. This not only reduces operational and maintenance costs but also leads to a substantial decrease in fuel consumption, thereby contributing to environmental sustainability.

Furthermore, BISMART's fuel consumption management system adds another layer of efficiency to mining operations. By proactively mitigating inconsistent fuel consumption through monitoring of fuel distribution, the system optimizes fueling operations. It additionally detects machines exhibiting irregular fuel consumption patterns, enabling prompt corrective measures and contributing to an overall enhancement in fuel efficiency.

Through the implementation of this system, we achieved a noteworthy reduction in fuel consumption from 0.62 to 0.56 liters per ton, equating to a commendable 9.7 percent decrease. Over the course of a year, this substantial achievement has led to a noteworthy reduction of approximately 3,000,000 kilograms of CO_2 emissions.

As previously explained, the rationale for reducing fuel consumption encompasses several key aspects. Firstly, it involves minimizing downtime and enhancing overall productivity through vigilant monitoring. Secondly, it includes the implementation of optimized strategies governing the allocation of trucks to loaders. Additionally, there is the regulation of truck velocities to maintain an optimal operational level. Furthermore, the fuel management system aids in planning fueling equipment, identifying machines with abnormal fuel consumption patterns, and facilitating timely maintenance.

5. Challenges and future directions

Efforts to establish sustainable mining practices is accompanied by certain challenges; in this section, we explore the key obstacles and potential future directions that can shape the industry's path toward greater environmental and operational efficiency.

Within the domain of mine planning, it is essential to acknowledge the individuality of each mine, accompanied by its unique challenges. Furthermore, the persistent presence of various uncertainties (geological uncertainty, equipment-based uncertainty, and economic uncertainty (Blom et al., 2019), such as weather conditions, fluctuations in cycle times, and the influence of drivers, continues to invigorate this field (Ozdemir and Kumral, 2019). Looking ahead, future directions should address these challenges while concurrently prioritizing advancements in existing methodologies through the integration of new technologies.

As previously highlighted, the literature consistently endeavors to identify alternative solutions for truck-shovel systems, aiming to mitigate fuel consumption and minimize environmental impacts, thereby fostering sustainable mining operations. However, these solutions are not devoid of challenges. Despite belt conveyors being perceived as the most energy-efficient method for material transportation in surface mines, their broader adoption is hindered by factors such as high initial investment costs, as well as limitations in terms of mobility and scalability for increasing belt capacities (Soofastaei et al., 2018). When considering alternative fuels, the widespread adoption of natural gas is constrained despite its notable potential for reducing fuel costs and emissions. This limitation arises from challenges such as decreased engine performance and significant increases in hydrocarbon and carbon monoxide emissions under specific engine operating conditions (Feng and Dong, 2019). Hydrogen's utilization as a fuel presents challenges rooted in production, storage, transportation, and supply complexities. Nevertheless, it offers potential as an additive to fossil fuels, serving as a feasible alternative. Major drawbacks include limited durability, susceptibility to high temperatures, CO_2 emissions, and substantial energy and production expenses (Figueiredo et al., 2023). In the context of CO_2 reduction, a trade-off exists between cost and the reduction of CO_2 emissions. In certain scenarios, it becomes challenging to find a solution that simultaneously minimizes both aspects (Kim and Kim, 2021).

To summarize, forthcoming directions must effectively address the identified challenges. Moreover, given the current global context and the exponential growth of generated data, it is imperative to increasingly employ cutting-edge technologies in the realms of AI and data science for the comprehensive analysis of this data.

Employing these tools for in-depth data analysis can provide valuable insights, enabling informed decision-making and reinforcing the commitment to sustainable practices within the mining sector.

6. Conclusions

The mining industry, characterized by its substantial energy consumption and substantial environmental impacts, is embracing the principles of sustainable mining to address these dual challenges. Energy conservation and environmental mitigation have become crucial priorities within the mining sector, with a particular emphasis on the loading and haulage processes due to their energy-intensive nature. This paper presents a comprehensive review of research papers, offering an integrated understanding of the influential factors in fuel consumption and subsequently delving into energy-saving and environmental protection strategies. The optimization strategies for reducing fossil fuel consumption primarily encompass three distinct avenues. Notably, a significant portion of attention has been directed towards refining mining short-term planning, especially in the context of truck-shovel allocation challenges, aiming to optimize fuel efficiency. Additionally, alternative methods like In-Pit Crushing and Conveying and fuel substitutes such as natural gas are introduced as means to enhance energy efficiency and decrease environmental impacts. Furthermore, maintenance plans are highlighted as potential solutions to reduce energy usage in mining fleet operations. However, these strategies are not devoid of challenges, and the literature strives to address them. For instance, in the realm of short-term planning, issues like the unique nature of individual mines and the persistent presence of various uncertainties persist. In the context of In-Pit Crushing and Conveying, challenges encompass initial investment costs and limitations regarding mobility and scalability when expanding belt capacities. Similarly, the adoption of natural gas as an alternative fuel presents obstacles, including reduced engine performance and significant hikes in hydrocarbon and carbon monoxide emissions. Thus, future research should focus on tackling the aforementioned challenges. Moreover, there exists a necessity to explore the application of advanced data analysis techniques to manage large volumes of generated data effectively, thereby improving decision-making processes.

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